



Statistical Analysis

(Significance, Hypothesis Testing)



Deepak Sharma
BSBE and DS&AI
IIT Roorkee



Sample, Population

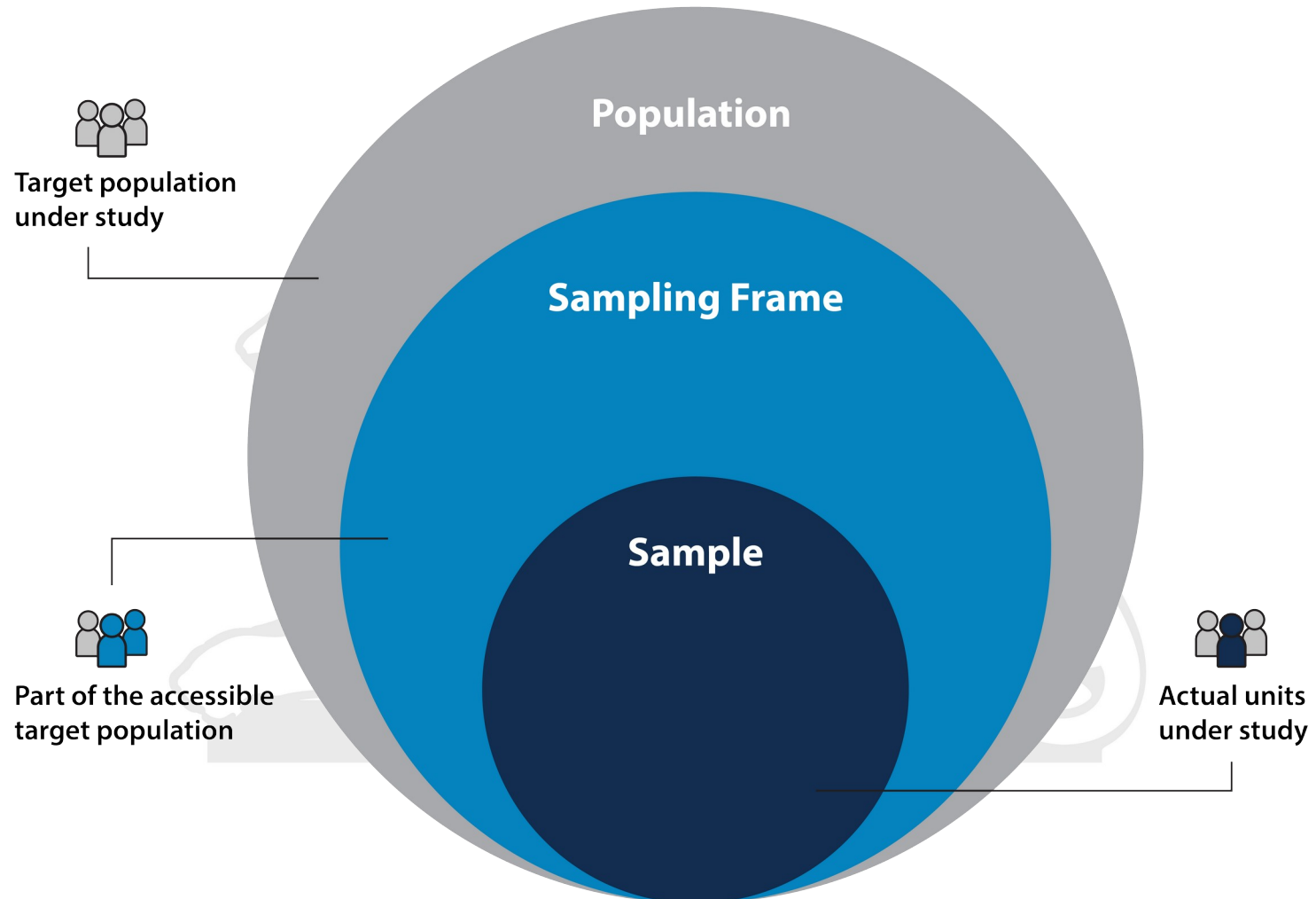
A **random sample** is a selection of some members of the population such that each member is independently chosen and has a known nonzero probability of being selected.

A **simple random sample** is a random sample in which each group member has the same probability of being selected.

The **reference, target, or study population** is the group we want to study. The random sample is selected from the study population.



Sampling



Parameter vs Static

Parameter

- A numerical value summarizing a characteristic of a population (e.g., population mean).

Statistic

- A numerical value summarizing a characteristic of a sample (e.g., sample mean).

Note: Parameter is fixed; Statistic varies across samples.

Example

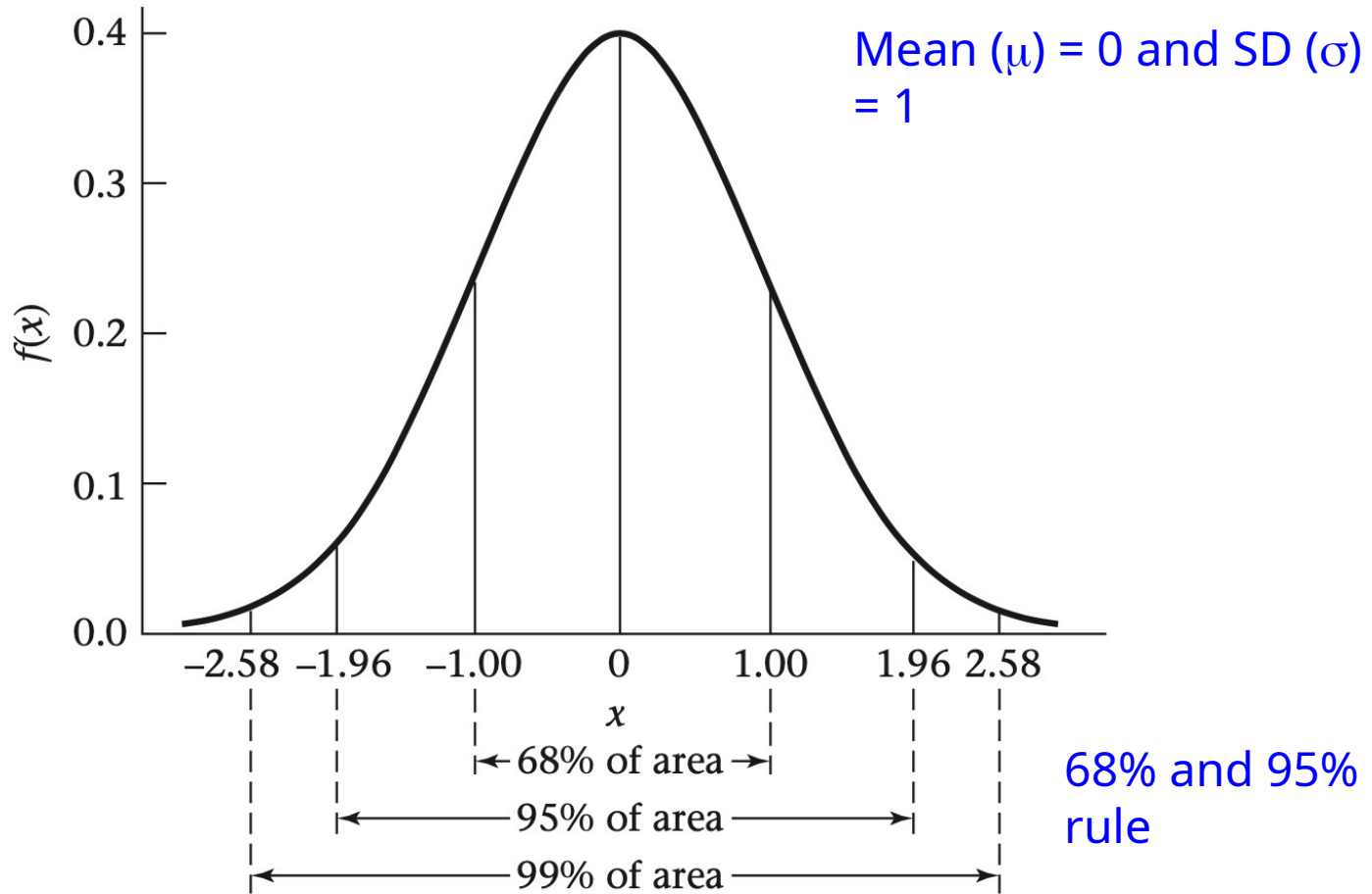
- Population mean height of all adults = 170 cm (**Parameter**).
- Mean height from a sample of 100 adults = 168 cm (**Statistic**).

Statistical significance

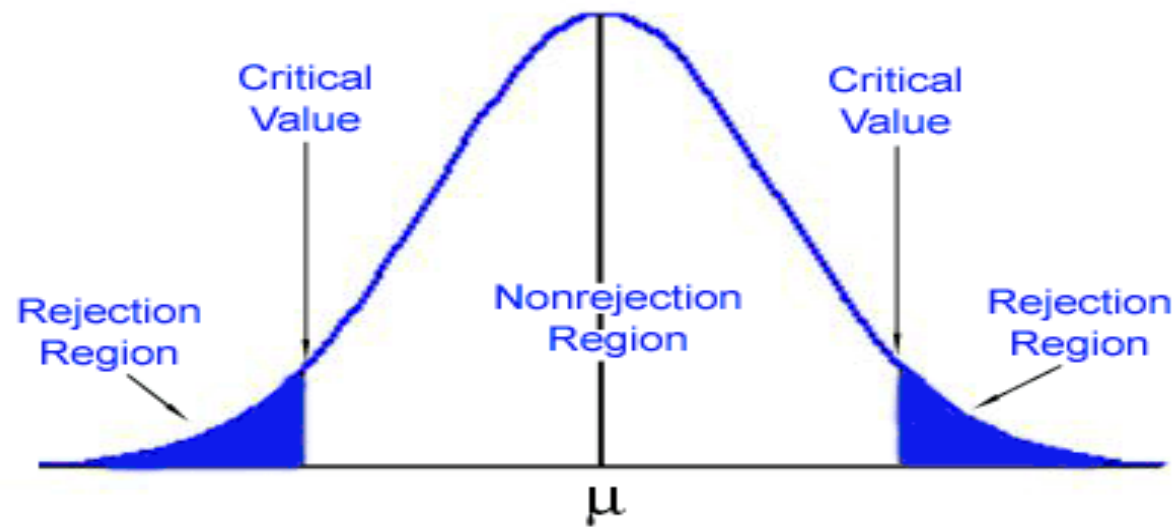


- An important aspect of sampling theory is the study of *tests of significance*, which enables us to decide:
 - A result is statistically significant i.e. it is unlikely to have occurred by chance alone, given a predefined significance level.
 - Statistical significance depends on the *p-value*.
 - *p* value is probability of obtaining test results at least as extreme as the observed results, assuming the null hypothesis is true.
 - *p*-values are calculated using different probability distributions depending on the test. A significant result is when the *p*-value is less than the chosen level of significance (usually 0.05).
Example: Evaluating whether a new algorithm performs better than the existing one.
- *Null Hypothesis (H_0)*: No difference in performance.
- *p*-value = 0.02 indicates evidence to reject the null hypothesis.

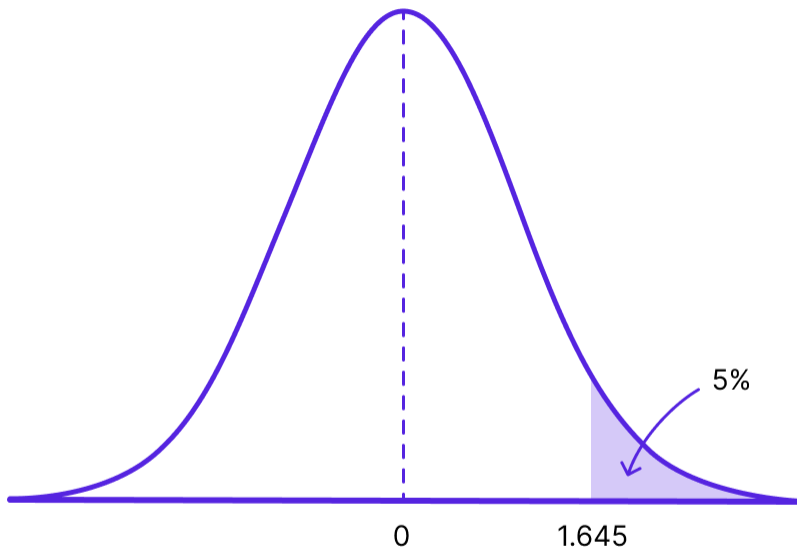
Standard normal distribution



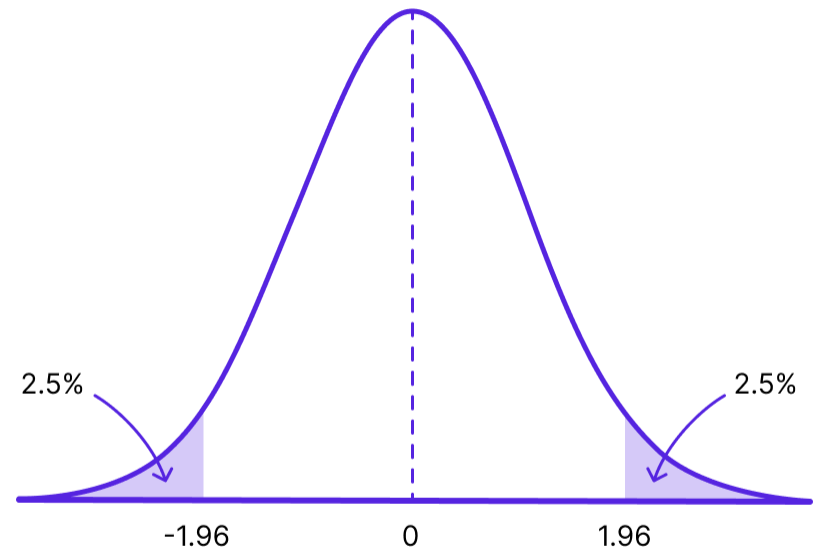
Hypothesis testing



Hypothesis testing



One-tailed test



Two-tailed test



Hypothesis testing

- *Method of making decisions or inferences about a population based on sample data*
- *For applying the test of significance, we first set up a hypothesis:*
 - *Null Hypothesis (H_0): Assumes no effect or difference.*
 - *Alternative Hypothesis (H_1): Assumes an effect or difference exists.*

Hypothesis testing, Error

Decision	Truth	
	H_0	H_1
Accept H_0	H_0 is true and H_0 is accepted	H_1 is true and H_0 is accepted Type II error
Reject H_0	H_0 is true and H_0 is rejected Type I error	H_1 is true and H_0 is rejected TB HI V

The probability of a **type I error** is the probability of rejecting the null hypothesis when H_0 is true.

The probability of a **type II error** is the probability of accepting the null hypothesis when H_1 is true.

Parametric vs. Non-parametric tests

Feature	Parametric test	Non-parametric test
Definition	Based on parameters (eg. Mean, variance, SD)	Not based on any parameter
Data distribution	Assumes normal/specific distribution	No assumption
Efficiency	More powerful if assumption holds	Less powerful but more flexible
Precision	More precise when assumption holds	Less precise but robust
Rule of thumb for use	Mean/2 > SD	Mean/2 < SD

Parametric tests

- *Involves inference about the population parameters.*

One sample test (One sided alternatives)

- *Suppose we want to test the hypothesis that mothers with low socio-economic status (SES) deliver babies whose birthweights are lower than “normal.”*
- *To test this hypothesis, a list is obtained of birthweights from 100 deliveries in a low-SES area. The mean birthweight ($\bar{x}$) is found to be 3.26 kg with a SD ($\bar{s}$) of 0.68 kg. We know that the national mean birthweight is 3.4 kg (μ_0). Can we say the underlying mean birthweight from low-SES area is lower than the national average?*

The **acceptance region** is the range of values of $\bar{x}$ for which H_0 is accepted.

The **rejection region** is the range of values of $\bar{x}$ for which H_0 is rejected.

A **one-tailed test** is a test in which the values of the parameter being studied (in this case μ) under the alternative hypothesis are allowed to be either greater than or less than the values of the parameter under the null hypothesis (μ_0), *but not both*.

One sample test (One sided alternatives)

- Suppose we want to test the hypothesis that mothers with low socio- economic status (SES) deliver babies whose birthweights are lower than “normal.”
- To test this hypothesis, a list is obtained of birthweights from 100 deliveries in a low-SES area. The mean birthweight ($\bar{x}$) is found to be 3.26 kg with a $\bar{x}/D (s)$ of 0.68 kg. We know that the national mean birthweight is 3.4 kg (μ_0). Can we say the underlying mean birthweight from low-SES area is lower than the national average?

One sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{3.26 - 3.4}{0.68/\sqrt{100}} = -2.05$$

$$t_{99, 0.05} = -1.66 \text{ (number of freedom, } n-1)$$

$t < -1.66$, **Reject H_0 (Different, lower)**

t distribution



Table 5 Percentage points of the t distribution ($t_{u,d}$)^a

Degrees of freedom, d	u								
	.75	.80	.85	.90	.95	.975	.99	.995	.9995
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657	636.619
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	31.598
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	12.924
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.767
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.551
60	0.679	0.848	1.046	1.296	1.671	2.000	2.390	2.660	3.460
120	0.677	0.845	1.041	1.289	1.658	1.980	2.358	2.617	3.373
∞	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.291

^aThe u th percentile of a t distribution with d degrees of freedom.

Source: Table 5 is taken from Table III of Fisher and Yates: "Statistical Tables for Biological, Agricultural and Medical Research," published by Longman Group Ltd., London (previously published by Oliver and Boyd Ltd., Edinburgh). Reprinted by permission of Pearson Education Ltd.

Non Parametric tests

- Not based on parameters (mean, variance).
- Sample size is small

Two sample Mann-Whitney test

- Compare observations from control (X) and treated (Y) groups
- H_0 : X and Y have same distribution
- H_1 : X and Y have different distribution

$$U = S_x - \frac{n_x(n_x + 1)}{2}$$

S_x : sum of ranks of x

n_x : number (sample size) of x

Mann Whitney test

The data shows the length of time (in days) that control mice (X) and inoculated mice (Y) live:

X : 20 23 28 31 30

Y : 20 29 23 48 41

H_0 : X and Y have same distribution

H_1 : X and Y have different distribution

	x	y	x	y	x	y	x	x	y	y
Arrange:	20	20	23	23	28	29	30	31	41	48
Ranks	1.5	1.5	3.5	3.5	5	6	7	8	9	10

$$U = 25 - \frac{5(5+1)}{2} = 10$$

Lower tail: 5; Upper tail: 20

Since $U = 10$ does not fall in critical region, null hypothesis (H_0) accepted

Mann Whitney test

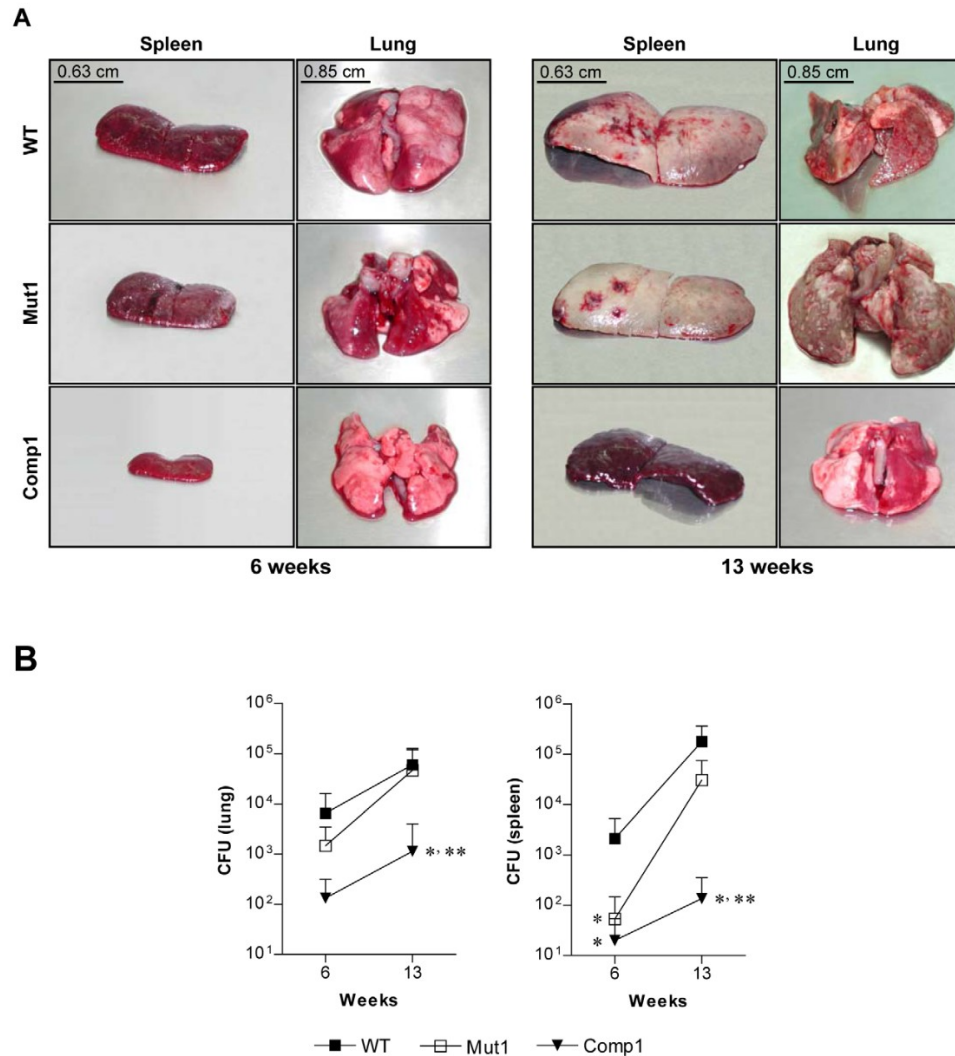


Figure 5. Virulence of passaged *M. tuberculosis* strains in guinea pigs. (A) Pictorial representation of lungs and spleens. **(B)** CFU in lungs and spleens are expressed as mean $\pm$ SD. *, ** represent $P < 0.05$ in comparison to WT and Mut1, respectively.
doi:10.1371/journal.pone.0009448.g005

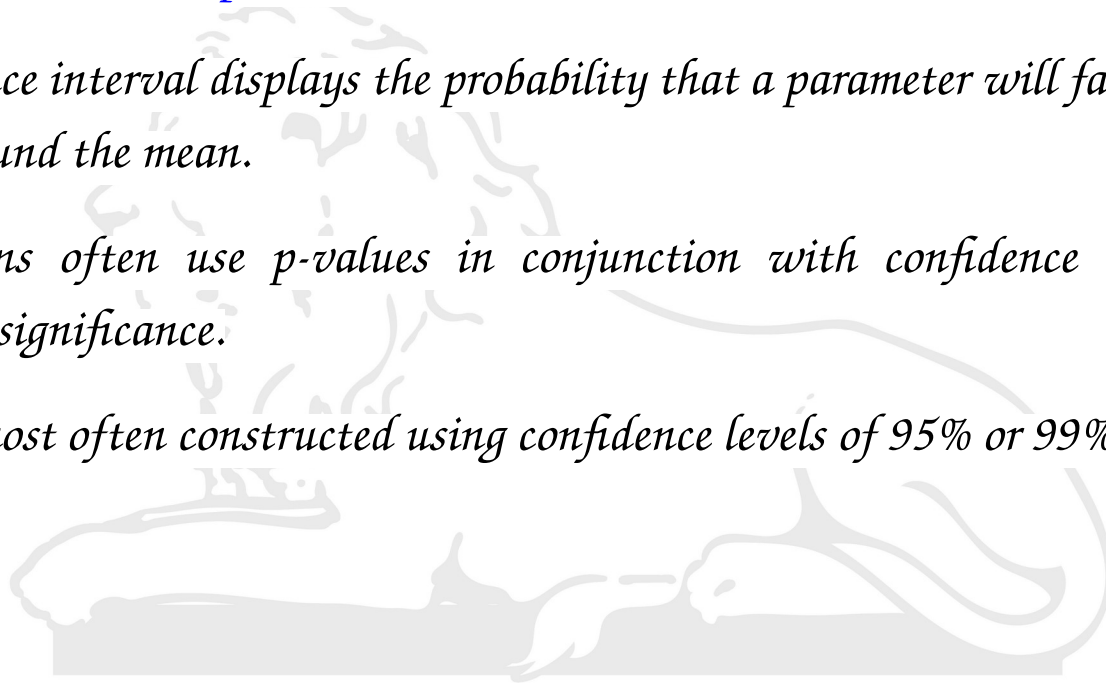
Majumdar SD, Sharma D *et al.*
(2010) *PLoS One*, 5, e9448.

Confidence Interval



It describes the variability surrounding the sample point estimate (the wider the interval, the less confident we can be about the estimate of the population mean). In general, the larger the sample size the better (more precise) the estimate is.

- A confidence interval displays the probability that a parameter will fall between a pair of values around the mean.*
- Statisticians often use p-values in conjunction with confidence intervals to gauge statistical significance.*
- They are most often constructed using confidence levels of 95% or 99%.*



Height of IITR students (100 samples; 5ft
 ± 1.5 ft)

CGPA of IITR students (100 samples; 7 ± 2)

Confidence Interval



The equation for Confidence Interval for the population mean when the population standard deviation is unknown and the sample size is large (over 30) is

$$CI = \bar{x} \pm z \frac{s}{\sqrt{n}}$$

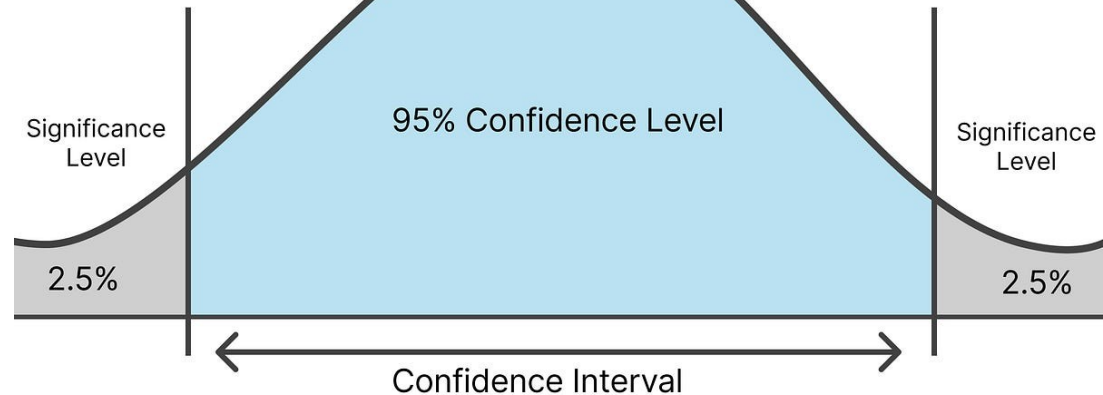
CI = confidence interval

$\bar{x}$ = sample mean

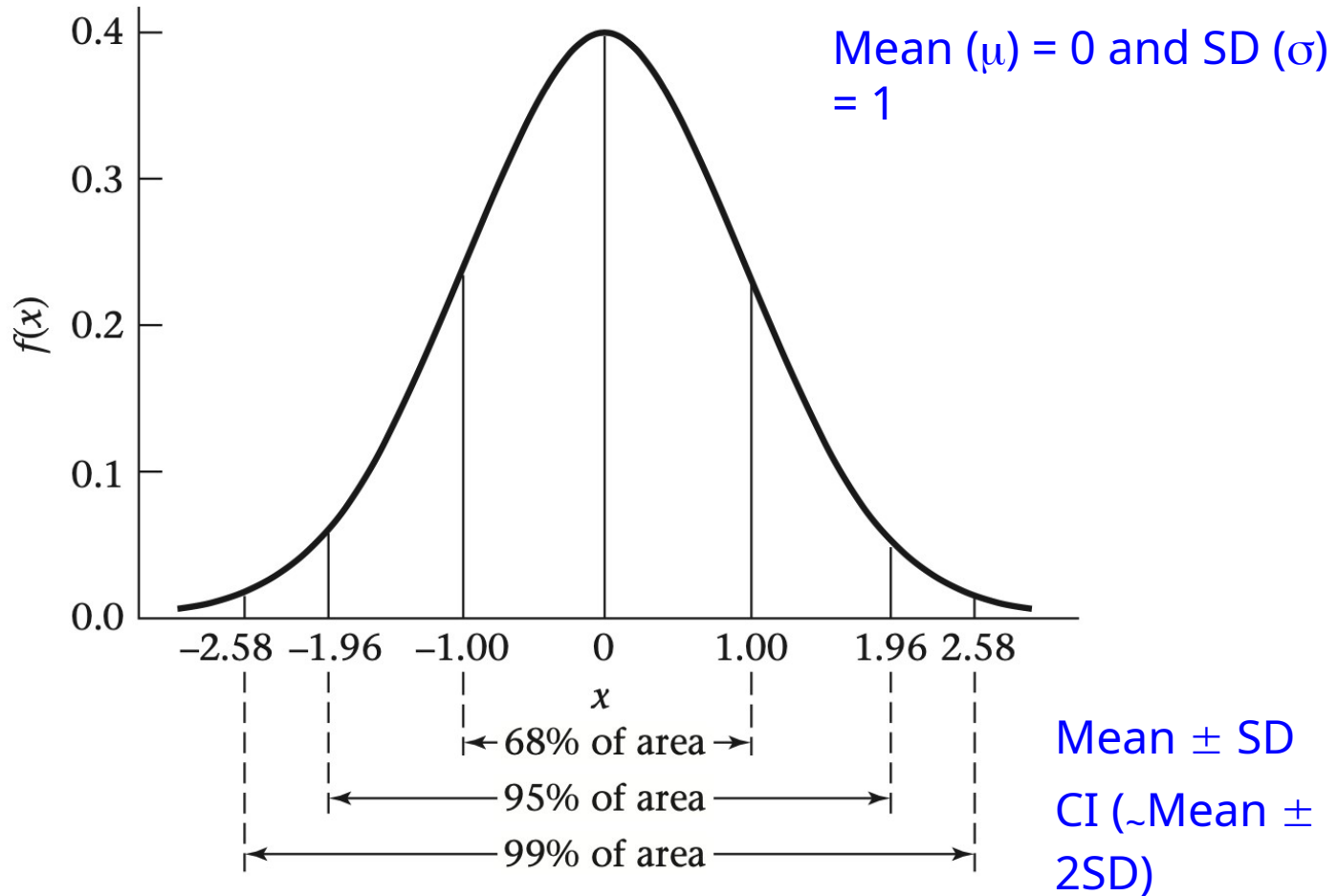
z = confidence level value

s = sample standard deviation

n = sample size



Standard normal distribution



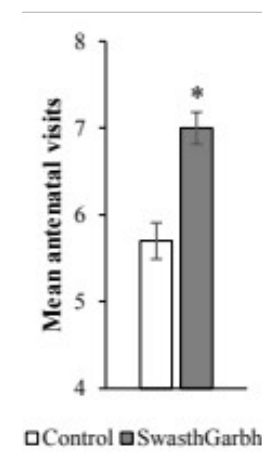


SwasthGarbh: Confidence Interval



Table S2: Relationship of treatment and socio-demographic variables on the number of ANC visits by regression analysis.

	Regression coefficient			
	Unadjusted (95% CI)	P-value	Adjusted (95% CI)	P-value
Mean difference in ANC visits	1.29 (0.74, 1.84)	<0.001	1.36 (0.82, 1.90)	<0.001
Age (years)	0.05 (-0.24, 0.12)	0.19		
POG at recruitment	-0.004 (0.13, 0.12)	0.95		
Higher Education	1.00 (0.43, 1.58)	0.001	0.35 (-0.31, 1.01)	
Lower Socio-economic Status	-1.06 (-1.71, -0.40)	0.002	-1.01 (-1.75, -0.27)	
Urban Area	-0.08 (-0.87, 0.71)	0.84		
Induction/IVF Conception	-0.22 (-0.92, 0.48)	0.54		
Gravidity >2	-0.23 (-0.86, 0.40)	0.48		
≥1 Abortions	-0.19 (-0.79, 0.41)	0.53		



Sharma S, Soni S *et al.* (2023) *IEEE JBHI*, 27, 2729-38.